ELSEVIER

Ecological Modelling 151 (2002) 51–61

ECOLOGICAL MODELLING

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'Hybrid' optimisation: a heuristic solution to the Markov-chain calibration problem

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Received 31 January 2001; received in revised form 3 October 2001; accepted 18 October 2001

Abstract

A Markov chain model for the natural course of succession through forest types was developed in a case study of mixed boreal forest located in Prioksko-Terrasnyi Biosphere Reserve, Near-Moscow Region, Russia. Stages of succession were specified in the conceptual scheme of succession transitions, and an original method of transition probability estimation, based upon expert data on the duration of the stages, generated a variety of transition matrices inheriting uncertainty both in the expert data and in the likelihood of alternative transitions in the scheme. To get rid of the uncertainty, the model has been calibrated to fit its observed state variables in terms of the relative area distribution among the specified forest types, the distribution becoming observable due to the application of a geographical information system (GIS) technology to forest inventory data. We formulate the calibration task as a multidimensional non-linear optimisation problem and find its formal approximate solution by means of a mathematical software routine. But the formal optimal solution has appeared to drive the model out of the expert-given intervals for stage durations, thus revealing disagreement between the model and observation data. Since the data may have errors, we have to look for a compromise in the model vs. data controversy. The formal solution, when combined with the original method of transition probability estimation, indicates the proper directions for a feasible choice of model parameters. Thus, the compromise solution (to the 'hybrid' optimisation problem) now consists in following those directions up (or down) to the nearest points on the expert-given intervals. Also, the formal solution has fixed the best likelihood ratios for the alternative transitions in the scheme, and the 'hybrid' inherits logically this advantage of the 'parent'.

Keywords: Succession; Markov chain; Transition probabilities; Calibration; Heuristics; Validation

1. Introduction

Model calibration, a fundamental notion of modelling in general and that of ecological modelling in particular, means finding 'the best accordance between computed and observed state

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PII: S0304-3800(01)00486-0

variables by variation of a number of parameters' (Jørgensen, 1986). Although some general and versatile recipes are known for the calibration procedure (ibidem), a particular type of models under study may predetermine the way in which the procedure applies.

In this regard, Markov chain models of vegetation succession (see e.g. Horn, 1975, 1981; Usher, 1979, 1981, 1992; Mirkin and Naumova, 1984; Acevedo et al., 1995; Logofet, 1999; Logofet and Lesnaya, 2000) feature the fact all the model parameters are gathered into a single matrix of transition probabilities, while the state variables constitute a finite probability distribution of vegetation types specified in the model. When the transition probabilities are estimated as frequencies of the corresponding transitions observed (see e.g. Usher, 1992; Vinogradov and Schitov, 1994; Li, 1995; Pomaz et al., 1998; Balzter, 2000), the model becomes pure statistical, hence calibration follows standard routes of statistical estimation. But when other considerations define construction of the model, calibration turns into a non-standard task, which is supposed to eventually remove uncertainties inherent in the model construction. This appears possible due to a general idea that, while the uncertainties generate a whole spectrum of potential model behaviours, the 'variation of a number of parameters' can make it certain if finds 'the best accordance between computed and observed state variables'. Here is a logical niche for optimal problem formulations, which are called to provide for 'automatic calibration' in the sense of Jørgensen (1986). However, the 'observed state variables' may not be absolutely true when the observation data bring errors. Consequently, the result of 'automatic calibration' may not have the first priority in the controversy between the original model and available data, and the task is to find a reasonable compromise to solve the controversy.

The present communication illustrates the aforesaid with a Markov chain model of succession in mixed boreal forest (Korotkov et al., 2001), which is constructed, under a given scheme of succession transitions, by the 'stage–duration' method of transition probability estimation suggested before (Logofet and Lesnaya, 2000). The 'inherent uncertainties' appear since expert estimates of stage durations vary within certain limits. While the characteristic time scale of forest successions expands normally to hundreds of years, data on the 'observed state variables' (in terms of the relative area distribution among forest types) have appeared to be available, due to current progress in geographical information system (GIS) technologies, at least at two time moments which are distant by almost 50 years. The calibration problem can thus be both mathematically posed and practically solved, with the practice breeding an idea of 'hybrid optimisation' that is presented in what follows below.

2. A Markov chain model of forest succession

The model to be calibrated describes forest succession on a watershed area in the coniferous-broadleaf forest subzone (Korotkov et al., 2001). Its conceptual scheme is presented in Fig. 1, which shows a natural (in the absence of severe disturbances) course of succession through forest types after a tillage area was abandoned—a scenario of forest secondary succession typical for the land-use history in the region. The numerals in boxes enumerate the stages of succession, which are specified in this way and supplied with expert estimates of the stage duration times.

In the mathematical sense, the model represents a discrete-time Markov chain of transitions among the 15 states, governed by a matrix of (time-homogeneous) transition probabilities, P, whose pattern mirrors the scheme, while quantitative entries determine the rates of the corresponding transitions. The *state variable* is therefore given by the stochastic vector, $\boldsymbol{x}(t)$, whose component $x_i(t)$ shows the probability that the chain is in state i ($i = 1, 2, \ldots, 15$) after t time steps ($t = 0, 1, 2, \ldots$). The model *parameters* are, respectively, the 19 one-step transition probabilities (corresponding to arrows of the scheme) which represent the non-zero off-diagonal entries into the matrix $P(1) = [p_{ij}]$. The diagonal entries are linked to them by the (row-)stochastic property of the matrix. The Markov property results in the well-known dynamic equation (see e.g. Roberts, 1976):

$$x(t+1) = x(t)P(1), \quad t = 0, 1, 2, \ldots, \tag{2.1}$$

with the general solution

$$x(t) = x(0)P(t) = x(0)P(1)^t, \quad t = 0, 1, 2, \ldots. \tag{2.2}$$

Given the matrix $P(1)$ and an initial value $x(0)$ of the state variable—identified usually with the relative area composition of the corresponding types in the mosaic of vegetation—formula (2.2) turns calculation of $x(t)$ into a matter of simple matrix multiplication. Thus, matrix $P(1)$ determines all the further properties of the model.

As regards the matrix, it is clear that if, for any pair of stage numbers $i \neq j$, there is no arrow $i \to j$ in the scheme, then $p_{ij} = 0$. To estimate non-zero p_{ij}, we use a simple method that is based on a fundamental property of finite Markov chains (Kemeny and Snell, 1960) and constructs the diagonal entries of $P(1)$ in a unique way (Logofet and Lesnaya, 2000): if m_i is the *average duration* given in the scheme for stage i (and expressed in time steps), then

$$p_{ii} = 1 - \frac{1}{m_i}, \quad i = 1, 2, \ldots, 14; \qquad p_{15,15} = 1. \tag{2.3}$$

If however the m_i values are only known to be within some ranges of variation, like those given in Fig. 1, then p_{ii} values vary in the corresponding ranges too, and this is the major source of uncertainties in the model construction. Additional uncertainty appears whenever there are *alternative* transitions in the scheme, i.e. $k \geq 2$ arrows $i \to j$

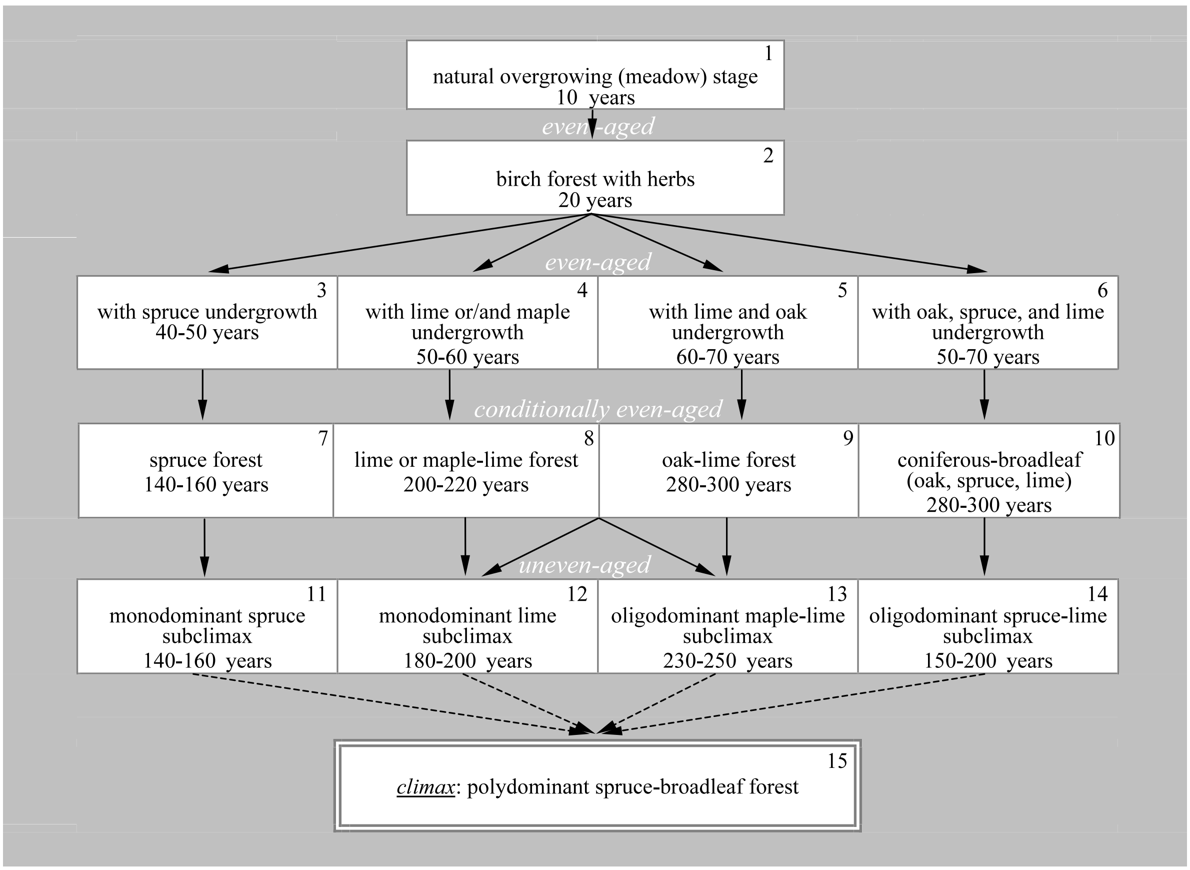

Fig. 1. Natural course of succession through forest types (after-tillage series) in Prioksko-Terrasnyi Biosphere Reserve, Near-Moscow Region, Russia (from Korotkov et al., 2001).

outgoing from state $i \neq j$, but no a priori knowledge of the ratios among the k values of p_{ij} (which must total to $1/m_i$ due to the stochastic property of matrix $P(1)$). This kind of uncertainty can be eliminated by the simplest assumption that all the transitions from state i are equiprobable, and thereafter

$$p_{ij} = \frac{1}{(km_i)}. \tag{2.4}$$

The assumption is not crucial for the method: any a priori knowledge of the likelihood ratio among such p_{ij} values can determine those values in a unique way (Logofet and Lesnaya, 2000).

If the time step is chosen to be $\Delta t = 5$ years, then the scheme in Fig. 1 suggests the following sets of m_i values:

$$\boldsymbol{m}_{\max} = [2 \quad 4 \quad 10 \quad 12 \quad 14 \quad 14 \quad 32 \quad 44 \quad 60 \quad 60 \quad 32 \quad 40 \quad 50 \quad 40], \tag{2.5}$$

$$\boldsymbol{m}_{\min} = [2 \quad 4 \quad 8 \quad 10 \quad 12 \quad 10 \quad 28 \quad 40 \quad 56 \quad 56 \quad 28 \quad 36 \quad 46 \quad 30], \tag{2.6}$$

with the components corresponding, respectively, to the minimal and maximal values of stage duration. The transition probability matrices $P_{\min}(1)$ and $P_{\max}(1)$ are then easily constructed by formulae (2.3) and (2.4), thus generating a corresponding range of variation in the model matrix. The pattern of model dynamic behaviour is qualitatively invariant on the whole range, though quantitative predictions of the model do depend on a particular choice within the range. A proper choice is supposed to become certain with solving the calibration problem.

3. Calibration of the transition matrix as an optimisation problem: formal solution

Application of a GIS technology (see Appendix A for some detail) to forest inventory data (collected in 1955 and 1999) enabled estimating the area of the main successional types enlisted in Fig. 1 (see Fig. 2 and Table 1). The lack of stages 11–15 can be explained both by a fairly young (in the successional time scale) age of the abandoned tillage area and by fairly strong anthropogenic influence on the natural course of forest succession (Korotkov et al., 2001). Moreover, even the potential to attain the climax state is highly doubtful in the current views of forest geobotany (ibidem).

The idea of calibration is simple: we take the 1955 data as the initial state vector $\boldsymbol{x}(0)$, project this vector, by Eq. (2.1) or Eq. (2.2), to the year of 2000 (in nine time steps if $\Delta t = 5$ years) and, taking the 1999 data as a reference vector (***Data***), we chose the uncertain parameters of the model in a way that minimises deviation of the model projection from the ***Data***.

Of the 19 independent model parameters, the value of p_{12} is fixed by the conceptual scheme, the values of $p_{11,15}, \ldots, p_{14,15}$ are put to zero due to the climax non-attainability conjecture, and hence the rest 14 parameters are to be optimised. As a measure of deviation, we use the Euclidean norm of vector difference, $\|\boldsymbol{x}_{2000} - \boldsymbol{Data}\|$, which totals errors through all the vector components, thus being similar to the popular 'least squares' of regression routines.

As a first approximation to an optimal solution, we have used matrix $P_{\min}$ corresponding to the case where all the m_i values are taken to be minimal and all the alternative transitions (i.e. $2 \to 3$, 4, 5, 6, $8 \to 12$, 13, and $9 \to 12$, 13) are assumed, in the absence of any relevant hypotheses, to be equiprobable with the competitor(s). It is not too surprising that deviation from the ***Data*** has appeared to be fairly great (Fig. 3a).

The next step is logically an attempt to fit the model through minimising the deviation (*goodness of fit*), which, in general, represents an *optimisation problem*. A number of mathematical questions arise here, in particular: 'Does there exist a matrix $P(1)$ such that

$$\boldsymbol{x}(0)P(1)^9 = \boldsymbol{Data}, \tag{3.1}$$

i.e. an *absolute solution* to the calibration problem?' To answer this question means to analyse what is called a *backward problem* in mathematics. Backward problems are generally known to be *ill-posed*, or *incorrect* in the mathematical sense. It means that either there is no unique solution to

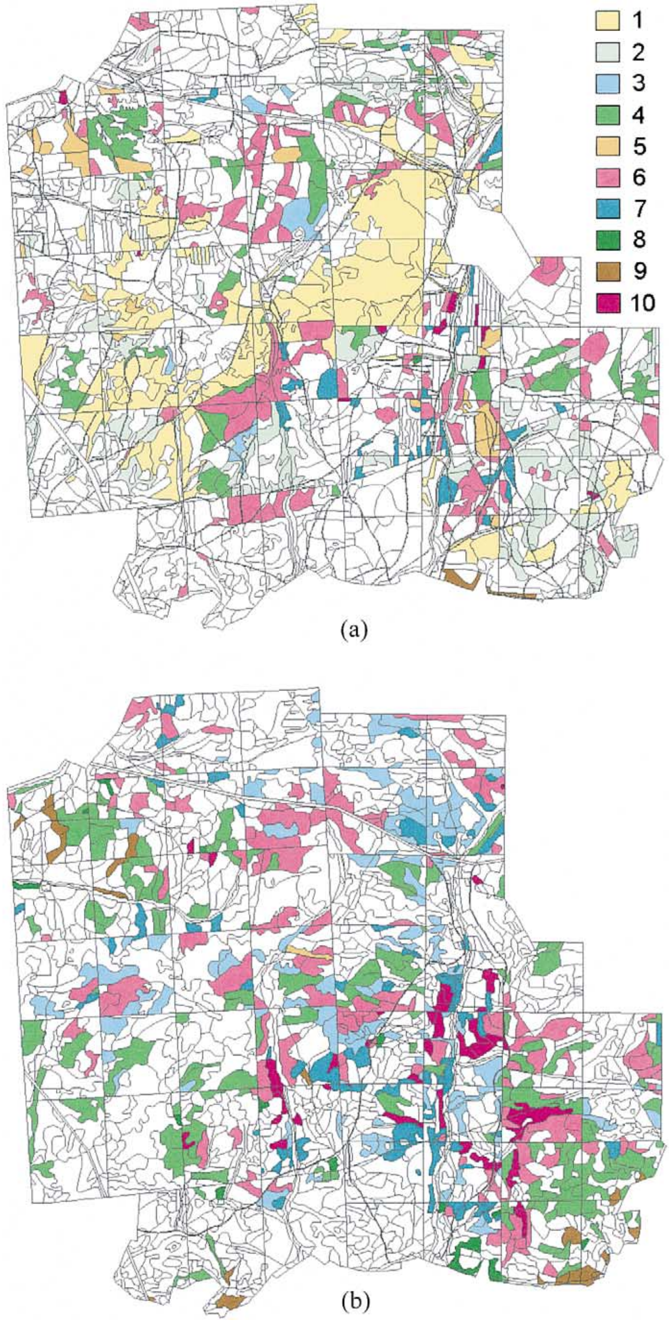

Fig. 2. Spatial distribution of the succession stages in Prioksko-Terrasnyi Biosphere Reserve (numerals in the legend correspond to the forest types enlisted in Fig. 1, white areas are those featuring strong influence of forest management before the Reserve foundation): (a) according to forest inventory data of 1954–1955; (b) the same of 1999.

Table 1
Relative areas of the successional types obtained from their spatial distribution in Prioksko-Terrasnyi Biosphere Reserve, near Moscow Region, Russia

Type number	Portion of the area occupied by a forest type (%)	
	1955 ($\boldsymbol{x}(0)$)	1999 (***Data***)
1	27.7	0
2	37.8	0
3	2.0	20.3
4	0.9	30.5
5	2.4	0.3
6	19.6	28.2
7	4.3	10.1
8	0.2	1.7
9	3.2	3.0
10	1.9	5.9
11	0	0
12	0	0
13	0	0
14	0	0
15	0	0
Total	100	100

the problem, or/and the solution lacks continuous dependence on the boundary conditions (vectors $\boldsymbol{x}(0)$ and ***Data*** in our case). In the model terms, it means that even small errors in data may cause drastic distinction between the models calibrated on those data.

When considered as a function of the 14 variables to be calibrated, the projecting operator $P(1)^9$ represents an essentially non-linear function. With only ten non-linear component equations involved actually in the vector-matrix Eq. (3.1), the backward problem of finding an exact calibration solution can be shown to be ill-posed. Therefore, it makes little sense to search for an exact solution, which, furthermore, would scarcely have any accurate meaning under inaccuracy of the data. Thus, we have rather to solve an optimisation problem:

$$\left\|\boldsymbol{x}(0)P(1)^9 - \boldsymbol{Data}\right\| \to \min_{\Omega}, \tag{3.2}$$

where Ω designates the 14D space of the transition probabilities to be calibrated. Problem (3.2) is a kind of non-linear multivariable optimisation problem, and modern systems of mathematical software offer routines for approximate solution of this kind of problem.

An approximate solution to problem (3.2), found by means of a (properly adjusted)

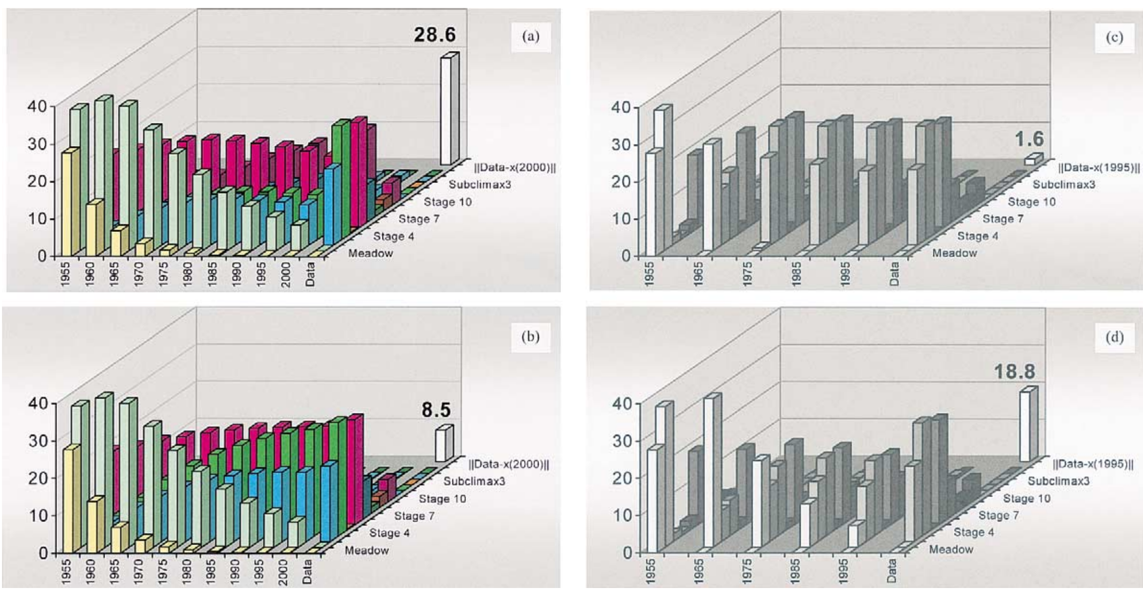

Fig. 3. Model dynamics in terms of the relative area distribution (%) among succession stages (a) before calibration; (b) after the formal calibration; (c) after the formal calibration for the doubled time step; (d) after the hybrid–heuristic calibration.

Table 2
Restoring stage duration values from the formally calibrated model

Stages of variable duration	2	3	4	5	6	7	8	9	10
Duration values (*years*)									
Expert maximum	20	50	60	70	70	160	220	300	300
Restored from diag(P)	10	↑	↑	44	↑	↑	↓	↓	↑
	↓	123	722	↓	283	Inf	122	123	1209
Expert minimum	20	40	50	60	50	140	200	280	280
Heuristic choice	20	50	60	60	70	160	200	280	300

MATLAB® 5.3 non-linear multivariable optimisation routine, yields a much better goodness of fit than does the first approximation (see Fig. 3b, averaging over 15 components would yield a less than 1% error estimation). Even better results are achieved if we double the time step to $\Delta t = 10$ years, hence halve the m_i values, and take operator $P(1)^4$ in problem (3.2) (Fig. 3c).

The high goodness of fit is not too surprising since we achieve it through solving a formal optimisation problem. The problem with formal solutions is however that a formal solution may happen to have no modelling sense, and this is exactly the case we face with.

4. Calibration of the transition matrix as a compromise problem: heuristic solution

Since the method we are using in model construction allows one to restore the stage duration data from the diagonal entries of a given matrix $P(1)$ in a unique way, we have done so for the calibrated matrix. The results shown in Table 2 are discouraging: the restored duration values have appeared to be far beyond their expert-given intervals ('Inf' denotes infinite time, i.e. the state becomes absorbing). It means that the result of formal calibration is inconsistent with the conceptual scheme of the model. But does it mean that the scheme ought to be revised? Not necessary. The scheme mirrors a generalised knowledge that vegetation science gained on this kind of forest succession, whereas the data may well have measurement errors inherent in the way the data are collected by forest inventories.

As an outcome of this contradiction, we suggest what might be called *hybrid-heuristic calibration*: although the formal optimisation solution is inconsistent, it prompts the right direction to a better fit, and the heuristics prompts using these 'recommendations' in further calibration efforts. It means that calibrated m_i values should represent now the feasible (by the expert-given intervals) choice closest to what is prompted by the formal calibration. This choice is made in the last row of Table 2, the choice directions being indicated by vertical arrows associated with the restored values.

Once the m_i values are fixed, the direct, original method can apply to find the diagonal entries of the transition matrix by formulae (Eq. (2.3)). But to complete this way of 'calibration' we still need to know certain ratios between the probabilities of alternative transitions. And this is what the formal calibration has already achieved since fixing each element of the matrix (see Table 3). With these ratios, we find a consistent model that has still an acceptable goodness of fit (Fig. 3d) as the accuracy of forest inventory data is commonly believed to be hardly better than 20%, especially in the 'pre-GIS age' of forestry (D.A. Starostenko, A.N. Filipchuk, personal communications). Therefore, tiny variations in the goodness of fit of an approximate optimal solution

Table 3
Likelihood ratios among the alternative transitions

	p_{23}:p_{24}:p_{25}:p_{26}	$p_{8,12}$:$p_{8,13}$	$p_{9,12}$:$p_{9,13}$
Before calibration	1:1:1:1	1:1	1:1
After calibration	36:46:0.0017:18	18:82	55:45

may have no real sense as far as no sense may be there in exact mathematical solution to the ill-posed problem (3.1). Those tiny variations did really occur in our manipulations with the MATLAB® optimisation routine, yet the ensuing heuristic choice (i.e. the last row of Table 2) has always been invariable.

5. Discussion

Since we do not know the absolute solution to the calibration problem and since the accuracy of data is known to be limited, it could hardly make sense to ask what is the accuracy of the hybrid-heuristic calibration. A sensible question however is whether the heuristics turn out rational, and a rational way to answer then consists in comparing the heuristic choice with non-heuristic ones. As a universal alternative to heuristics, we consider random choice of the model parameters within the expert-given intervals. A series of 2000 direct Monte-Carlo experiments on random construction of matrix $P(1)$ (Forest Successions, 2000) has revealed the random 'calibration' to be although competitive with the deliberate first approximation, but certainly worse than the heuristic calibration: even the minimal deviation (3.2) recorded in the whole series (namely, 27.4%) has appeared to be markedly worse than the goodness of fit achieved by heuristics (Fig. 3d). The later thus means the best compromise for the model–data controversy we faced with.

What the formal calibration has prompted on the likelihood ratios for the alternative transitions (Table 3) deserves special attention. A 5% shift in the $p_{9,12}$: $p_{9,13}$ a priori balance can hardly give a reason for any speculation in view of what has been said at the end of the previous section. But a marked preference in likelihood of transition $8 \rightarrow 13$ over that of $8 \rightarrow 12$ means that maple renewal is certainly more likely than maple extinction in this course of succession. At the lack of any priorities in the first approximation, transition $2 \rightarrow 5$ is unexpectedly 'recommended' to be a 1000 times less likely than the alternative ones.

An immediate explanation to the latter misbalance can, however, be found in the history of Reserve management. While elimination of the early succession stages and a respective increase in the area of the subsequent stages (Table 1) conform to the general course of succession, the drastic decline of forest type 5 (birch forest with lime and oak undergrowth) can be explained by rapid increase in population density of ungulates, such as aboriginal European elk (*Alces alces* L.), and introduced roe deer (*Capreolus capreolus* L.), red deer (*Cervus elaphus* L.) and sika deer (*Cervus nippon hortulorum* Swinhoe), which was observed in the Reserve in 1950–1980 as a result of the protection regime in combination with special biotechnical measures (Alexandrova, 1957; Zablotskaya, 1979; Alferova and Pererva, 1991). Ungulates rendered considerable pressure on forest vegetation, browsing the young growth of oak with the highest preference, but scarcely the lime and spruce growth if not at all (Yungerson, 1971; Rusanov and Sorokina, 1984). This makes it practically improbable that the forest succession went in the $2 \rightarrow 5$ direction those years, and model calibration has just confirmed this implication.

Meanwhile, in 1990s, the numbers of ungulates were observed to stabilise after some decline due to self-regulation effects and regulation efforts (Knyaz'kov, 2000a,b; M.N. Brynskikh, personal communication). An ensuing attenuation of the browsing pressure should, respectively, smooth the current misbalance in the likelihood ratio when the model is used to project the future course of succession in the Reserve. So, the ratios can hardly remain constant within any long projection period, thus giving another illustration to the homogeneity/inhomogeneity dilemma in Markov chain modelling discussed before (Logofet et al., 1997; Logofet and Denisenko, 1999).

A curious observation concerns improvement in the goodness of fit after doubling the time step (see Fig. 3b and c). Everyone who ever took courses on computational mathematics has been educated by the idea that this is rather making the time step smaller which provides for better approximation results. This common view has, however, grown from the discrete-time (difference) representation of originally continuous (differential) models. Since our model, on the contrary, originates from a deliberate discrete-time view of the succession process, its contradicting to the common perception should not be surprising. Moreover, the observation reveals that a proper value of the time step may also be a target for optimisation among the discrete Markov chain models. It also contributes to the general stipulation which the present case study is introduced with: '…a particular type of models under study may predetermine the way in which the (calibration) procedure applies'.

In general, after a model has been calibrated, the next canonical step should be its *validation*, which has 'to test selected parameters with an independent set of data' (Jørgensen, 1986). In our case, however, testing the model versus 'an independent set of data' can only mean projections to the 21st century and has to be postponed until new forest inventory data appear in 5, 10, or more years from now. The future projections, as follows from the above paragraph, can no longer preserve certain likelihood ratios fixed by calibration as the ratios are associated with a particular management scenario in the Reserve and are to vary together with variations in the scenario. Therefore, these model parameters, although having been calibrated, must nevertheless be excluded from the canonical independence between the calibration and validation procedures. This 'canon breakage' is perhaps a 'pay' for leaving the particular uncertainty in model construction to the general scope of model calibration, and this kind of observation may be apparently true for any kind of time-inhomogeneous Markov chain model.

6. Conclusion

The calibration problem for a Markov chain model of forest succession is practically solved due to application of a GIS technology to forest inventory data. Heuristic 'hybridisation' between the formal calibration procedure and the original method of transition matrix construction has resulted in a calibrated model that is consistent with the conceptual scheme of succession and the expert-given intervals for the duration of its specified stages. With potential errors in the observation data, the heuristic solution thus signifies the best compromise to the model–data controversy revealed by formal calibration. Validation, as the next logical step in model development, has to wait for the next forest inventory data, while some calibrated parameters will have still to be modified to be in conformity with a future scenario of reserve management.

Acknowledgements

INCO-Copernicus Programme of the EC supports the research, project No. IC15-CT98-0104. Calculations and graphics are implemented by means of MATLAB® software, Version 5.3. We would be grateful to any reader who could substitute 'an electronic copy' for 'a reprint' in his/her standard reprint-inquiring postcard.

Appendix A

The cartographic database for Prioksko-Terrasnyi Biosphere Reserve has been developed by means of TOPOL®, a universal software tool to create geo-information systems. TOPOL® enables one to work both with raster data and with cartographic vector data, associating the data with semantic information not only locally but also as external databases (Greshnov and Starostenko, 1999; Starostenko, 2000). Cartographical materials are all presented in the uniform system of Gauss–Kruger coordinates. Topographic maps (scaled 1:10 000) provide the cartographic basis (the raster layer), and the vector layer with forest strata boundaries (at the same scale) is referred to the basis. Forest maps are correspondingly related with an accuracy of 10–20 m (Starostenko, D.A., personal communication). Each polygon (forest stratum) has an associated external database with characteristics of the forest ecosystem state, including data on the tree, undergrowth, and underwood layers and the herb cover. The databases are compiled of data from a number of forest inventories and complemented with data gained in geobotanic field studies.

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